

Rubber Band Helicopter

(Coach Scheduled Event - 45 Minutes)



Description: Contestants will build and test one rubber band helicopter of any design using the materials provided at the competition, as well as take a written test on the principals of rotary flight.

Participants per assigned Team Number: 2

- If your school has 1 team you will send 2 students
- If your school has 2 teams you will send 2 students per team number; team numbers may not intermix
- If your school has 3 teams you will send 2 students per team number; team numbers may not intermix

Safety Equipment - Teams Must Bring:

- **Safety glasses labeled ANSI Z87+ (impact rated)**
- **All competitors must wear their eye protection during any competitor's test phase of the competition.**
- **If a team does not have the required eye protection, they will be given the opportunity to obtain it, time allowing, but will not receive extra time.**
- **If a team is unable to obtain eye protection, the team will not compete and will receive a no-show score.**

Impact Safety Glasses Required



Eyeglasses are not safety rated!

Team members must also bring:

- Pencils,
- A ruler/straight edge
- Scissors
- A team chart showing testing results for flying a practice helicopter on an 8 ½ in x 11 inch sheet of paper. Minimum requirement: 6 tests with different rubber band sizes, with and without weights.

Materials Provided at Event:

- 1 ≈ 15cm (6 in) Propeller
- 1 ≈ 15cm (6 in) Craft Stick
- 1-Jumbo Paperclip 9 cm (3.5 in)
- 2- Rubber bands – student choice from an assortment of sizes #64, #33, and #19
- 1-Sheet of Card stock ≈22 cm by 30 cm (8 ½ in x 11 in).
- 50 cm of 2.54 cm (1 in) masking tape
- 10 grams (0.35 oz) of plasticine
- Tin split shot fishing weight - .05 ounce
- Pliers

THE COMPETITION

Written Assessment: The team members will take a test on the principles of rotary flight.

As part of the written test, the team will estimate the time their helicopter will stay aloft without touching the highest point in the launch area.

Construction Phase (15 min):

- Teams will build their helicopter using any of the materials provided. They do not have to use all of the materials provided.
- If the team chooses to use the weights, a pair of pliers will be available.
- Rotation of the helicopter (clockwise, counterclockwise, horizontal, or vertical) is a team choice.
- The height of the ceiling will be announced during the building phase.
- Contestants may test their devices in the building area, and at the discretion of the event supervisor, test them from the official launch location.
- Teams must write their team number on their helicopter.
- During the construction phase the team will give the Event Supervisor their estimate for the time that their helicopter will stay airborne without touching the highest point of the launch area.

Helicopter Testing:

- Each team will have two launches of their helicopter.
- Maximum height will depend on the launch area ceiling or obstruction.
- All teams will release their helicopter from a height 1 meter (3.3 ft) from the floor of the launch area.
- Participants will announce their intention to launch their helicopter by saying “3, 2, 1, Launch.”
- Time will start when judges see the helicopter leave the hand of the student.
- The Event Supervisors will record the time a team’s helicopter stays airborne without touching the top of the launch area or any obstruction.
- Timing will continue until the helicopter touches the top of the launch area, an obstruction, or lands on the floor.
- The Event Supervisor will record and average the flight times for both launches. Times will be recorded in seconds to the nearest 1/100.

Scoring:

Final Score (FS) = Assessment Score + Time Score, max = 100

- Assessment Score $(TAS \div CAS) \times 50$; rounded to the 100th place value; maximum 50 points
 - Team Assessment Score (TAS) = the team’s score on the written assessment
 - Competition Assessment Score (CAS) = the highest score on the written test for all teams
- Time Score $(CTS \div TTS) \times 50$; rounded to the 100th place value; maximum 50 points
 - Competition Time Score (CTS) = the average longest time of all teams
 - Team Time Score (TTS) = the average time the team’s helicopter stays in the air
- Example
 - Highest Assessment Score across teams (CAS) = 15.
 - The team scores 9 on the assessment (TAS).
 - Assessment Score for the team = $(9 \div 15) \times 50 = 30$ points
 - Best Flight Time across teams (CTS) = 1.45 seconds.
 - The team’s flight time is 3.56 (TTS).
 - Time Score for the team = $(1.45 \div 3.56) \times 50 = 20.36$ points
 - **Final Score (FS)** = $30 + 20.36 = 50.36$ points

Tiebreakers:

- 1st - The closest estimate to the actual flight time.
- 2nd - Completeness of the team chart showing testing. Teams showing more testing details will be ranked above others.
- 3rd - Written Test Score

Resources:

- [Instuctables: How to make a helicopter for kids out of a coffee cup.](#)
- [University of West Virginia: Rubber Band Helicopter - STEMCARE Lesson](#)
- [Rubber Band Helicopters](#)
- Science Teachers & Librarians
- Get permission from your parents/coach to use AI to create a study guide using the rules provided here

Allowed Calculator

This is the acceptable calculators for all Division A events.

Class I - Stand-alone non-graphing, non-programmable, non-scientific 4-function or 5-function calculators. These are the most basic types of calculators. They are limited to the four basic mathematical functions and square roots. They are often cheap, costing as low as a dollar online.



Safety Glasses

Indirect Vent Goggles or Safety glasses are a mandatory safety requirement for some events. It is important to choose the type of protection specific to the task. Competitors are responsible for providing their own protective eyewear. Teams will not be allowed to compete without adequate eye protection.

Category C: Indirect Vent Goggles



Category B: Safety glasses



Compliant Eyewear Categories:

CATEGORY B

- Description: Impact protection. They provide protection from a high inertia particle hazard (high mass or velocity)
- Corresponding ANSI designation/required marking: Z87+.
- Category B eyewear can look either like goggles or safety glasses, but must have the Z87+ marking embossed on them.

CATEGORY C

- Description: Indirect vent chemical/splash protection goggles. These seal tightly to the face completely around the eyes and have indirect vents constructed so that liquids do not have a direct path into the eye (or no vents at all). If you can see through the vent holes from one side to the other, they are NOT indirect vents
- Corresponding ANSI designation/required marking: Z87 (followed by D3 is the most modern designation but, it is not a requirement)

Examples of Non-Compliant Eyewear:

- Face shields/visors are secondary protective devices and are not approved.
- Prescription Glasses containing safety glass should not be confused with safety spectacles. "Safety glass" indicates the glass is made to minimize shattering when it breaks. Unless these glasses bear the Z87+ mark they are not approved for use.

Notes:

1. A goggle that bears the Z87+ mark and is an indirect vent chemical/splash protection goggle will qualify for both Category B & C
2. VisorGogs do not seal completely to the face, but are acceptable as Type C goggles